Formula Sheet Part 4: GARCH, Risk Forecasting, and Exam Templates

This is a two-column topic split from `All_Formulas_From_Chatnotes.md`.

Conventions

$$t, T, h, k, n, p, q, N$$

Variables: t is a time index; T is sample size or the final observed time; h is a forecast horizon; k is a lag or number of model parameters depending on context; n is a return horizon; p, q are lag orders; N is the number of assets.

$$F_t$$

Variables: F_t is the information set available at time t.

Returns may be written as decimals or percentages. Check the scale before applying VaR or standard deviation formulas.

GARCH, Forecasting Volatility, and Model Comparison

ARCH(q) variance equation

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \cdots + \alpha_q \epsilon_{t-q}^2$$

Variables: this is the pure ARCH variance equation.

GARCH(p,q)

$$\begin{aligned} \epsilon_t &= \sigma_t u_t, \qquad u_t \sim \text{iid}(0,1) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \cdots + \alpha_q \epsilon_{t-q}^2 \\ &\quad + \beta_1 \sigma_{t-1}^2 + \cdots + \beta_p \sigma_{t-p}^2 \end{aligned}$$

Variables: β_j are coefficients on lagged conditional variances; p is GARCH order; q is ARCH order.

GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: α_1 is the shock reaction; β_1 is volatility persistence from lagged variance.

GARCH(1,1) conditions

$$\alpha_0 > 0, \qquad \alpha_1 \geq 0, \qquad \beta_1 \geq 0, \qquad \alpha_1 + \beta_1 < 1$$

Variables: $\alpha_1 + \beta_1 < 1$ gives finite unconditional variance and mean reversion.

GARCH persistence

$$\rho = \alpha_1 + \beta_1$$

Variables: ρ measures persistence of volatility shocks.

GARCH long-run variance and volatility

$$\text{Var}(\epsilon_t) = \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \qquad \bar{\sigma} = \sqrt{\frac{\alpha_0}{1 - \alpha_1 - \beta_1}}$$

Variables: $\bar{\sigma}^2$ is long-run variance; $\bar{\sigma}$ is long-run volatility.

GARCH conditional and unconditional moments

$$\begin{aligned} \mathbb{E}(\epsilon_t \mid F_{t-1}) &= 0 \\ \text{Var}(\epsilon_t \mid F_{t-1}) &= \sigma_t^2 \\ \text{Cov}(\epsilon_t, \epsilon_{t-k} \mid F_{t-1}) &= 0 \end{aligned}$$

Variables: conditional variance changes over time even when conditional mean is zero.

$$\begin{aligned} \mathbb{E}(\epsilon_t) &= 0 \\ \text{Cov}(\epsilon_t, \epsilon_{t-k}) &= 0 \\ \text{Var}(\epsilon_t) &= \frac{\alpha_0}{1 - \alpha_1 - \beta_1} \end{aligned}$$

Variables: standard GARCH errors are uncorrelated but not independent in squares.

Normal GARCH innovation skewness

$$u_t \sim \text{iid}\mathcal{N}(0,1), \qquad \epsilon_t \mid F_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$$

Variables: conditional normality makes conditional skewness zero.

$$\mathbb{E}(\epsilon_t^3 \mid F_{t-1}) = 0, \qquad \mathbb{E}(\epsilon_t^3) = \mathbb{E}[\mathbb{E}(\epsilon_t^3 \mid F_{t-1})] = 0$$

Variables: the unconditional third moment is zero under symmetric normal innovations.

$$\text{skewness} = \frac{\mathbb{E}(\epsilon_t^3)}{[\text{Var}(\epsilon_t)]^{3/2}} = 0$$

Variables: skewness is zero for standard symmetric GARCH innovations.

Mean equation plus GARCH volatility equation

$$\begin{aligned} r_t &= c + \phi_1 r_{t-1} + \epsilon_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

Variables: c, ϕ_1 define the mean equation; the second line defines volatility.

Constant-mean return equation

$$r_t = \mu + \epsilon_t$$

Variables: μ is constant mean return.

Mean-adjusted AR form

$$r_t - \mu = \phi(r_{t-1} - \mu) + \epsilon_t$$

Variables: ϕ is the AR coefficient around mean μ.

$$r_t = c + \phi r_{t-1} + \epsilon_t, \qquad c = \mu(1 - \phi)$$

Variables: c is the equivalent intercept.

$$r_t = \mu + \phi(r_{t-1} - \mu) + \epsilon_t = \mu(1 - \phi) + \phi r_{t-1} + \epsilon_t$$

Variables: these are algebraically equivalent AR(1) mean forms.

AR(1)-GARCH(1,1) one-step mean forecast

$$\mathbb{E}(r_{T+1} \mid F_T) = \hat{r}_{T+1} = c + \phi r_T$$

Variables: r_T is the latest known return.

GARCH one-step variance forecast

$$\hat{\sigma}^2_{T+1} = \mathbb{E}(\sigma^2_{T+1} \mid F_T) = \alpha_0 + \alpha_1\epsilon^2_T + \beta_1\sigma^2_T$$

Variables: ϵ^2_T is the latest squared shock; σ^2_T is the latest conditional variance.

Prediction interval using forecast variance

$$r_{T+1} \mid F_T \sim \mathcal{N}(\text{mean forecast}, \text{variance forecast})$$

Variables: mean forecast is $\hat{r}_{T+1}$; variance forecast is $\hat{\sigma}^2_{T+1}$.

$$\text{mean forecast} \pm 1.96\sqrt{\text{variance forecast}}$$

Variables: 1.96 is the approximate 95 percent normal cutoff.

Likelihood-ratio comparison of ARCH and GARCH

$$LR = 2(LL_{GARCH} - LL_{ARCH}) = 2(\ell_1 - \ell_0)$$

Variables: LL_{GARCH} is the unrestricted log likelihood; LL_{ARCH} is the restricted log likelihood.

$$LR \sim \chi^2_p$$

Variables: p is the number of restrictions or extra parameters tested.

Model comparison rules

$$\text{higher log likelihood} \Rightarrow \text{better}$$
$$\text{lower AIC/BIC} \Rightarrow \text{better}$$
$$\text{squared standardised residuals with no autocorrelation} \Rightarrow \text{better diagnostics}$$

Variables: these compare fitted volatility models.

Asymmetric Volatility Models

Standard GARCH symmetry

$$\sigma^2_t = \alpha_0 + \alpha_1\epsilon^2_{t-1} + \beta_1\sigma^2_{t-1}$$

Variables: squaring removes the sign of the shock.

$$(+a)^2 = (-a)^2 = a^2$$

Variables: a is any shock size.

GJR-GARCH model

$$\begin{aligned} r_t &= \mu + \epsilon_t \\ \epsilon_t &= \sigma_t u_t \\ \sigma^2_t &= \alpha_0 + \alpha_1\epsilon^2_{t-1} + \lambda I_{t-1}\epsilon^2_{t-1} + \beta_1\sigma^2_{t-1} \end{aligned}$$

Variables: λ is the leverage/asymmetry coefficient; I_{t-1} is a bad-news indicator.

GJR indicator

$$I_{t-1} = \begin{cases} 1, & \epsilon_{t-1} \le 0 \\ 0, & \text{otherwise} \end{cases}$$

Variables: the indicator switches on for non-positive shocks.

GJR good-news and bad-news effects

$$\begin{aligned} \epsilon_{t-1} > 0 : &\quad \text{effect} = \alpha_1\epsilon^2_{t-1} \\ \epsilon_{t-1} \le 0 : &\quad \text{effect} = (\alpha_1 + \lambda)\epsilon^2_{t-1} \end{aligned}$$

Variables: good news uses only α_1; bad news uses $\alpha_1 + \lambda$.

$$\text{good news effect} = \alpha_1, \qquad \text{bad news effect} = \alpha_1 + \lambda$$

Variables: a positive λ means bad news has a larger variance effect in GJR.

GJR leverage test

$$\begin{aligned} H_0 &: \lambda = 0 \\ H_1 &: \lambda \neq 0 \\ t &= \frac{\hat{\lambda}}{\text{se}(\hat{\lambda})} \end{aligned}$$

Variables: $\hat{\lambda}$ is the estimated leverage coefficient.

$$|t| > 1.96 \Rightarrow \text{reject } H_0 \text{ at approximately 5 percent}$$

Variables: rejecting means statistically significant asymmetry.

GJR parameter conditions

$$\alpha_0 > 0, \qquad \alpha_1 \ge 0, \qquad \beta_1 \ge 0, \qquad \alpha_1 + \lambda \ge 0$$

Variables: these keep conditional variance non-negative for good and bad news.

GJR unconditional variance

$$\text{Var}(\epsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1 - \lambda/2}$$

Variables: $\lambda/2$ appears when positive and negative shocks are equally likely.

EGARCH model

$$\log(\sigma^2_t) = \omega + \alpha_1 u_{t-1} + \gamma_1(|u_{t-1}| - \mathbb{E}|u_{t-1}|) + \beta_1 \log(\sigma^2_{t-1})$$

Variables: ω is intercept; α_1 captures signed shock asymmetry; γ_1 captures magnitude effects; β_1 captures log-volatility persistence.

EGARCH standardised shock

$$u_{t-1} = \frac{\epsilon_{t-1}}{\sigma_{t-1}}$$

Variables: u_{t-1} standardises the shock by conditional volatility.

Expected absolute standard normal shock

$$\mathbb{E}|u_t| = \sqrt{\frac{2}{\pi}} \approx 0.7979$$

Variables: this value is used when u_t is standard normal.

EGARCH positivity

$$\log(\sigma^2_t) \text{ is modelled } \Rightarrow \sigma^2_t \text{ is automatically positive}$$

Variables: modelling log variance avoids non-negativity constraints on coefficients.

EGARCH leverage sign in the notes

$$\lambda < 0 \Rightarrow \text{negative shocks raise volatility more than positive shocks}$$

Variables: λ is the asymmetry coefficient in the note's EGARCH interpretation.

News impact curve for ARCH(1)

$$NIC(\epsilon_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Variables: NIC maps yesterday's shock into today's conditional variance.

News impact curve for GARCH(1,1)

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: $\bar{\sigma}^2$ is the variance level used to hold lagged variance fixed.

$$NIC(\epsilon_{t-1}) = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \bar{\sigma}^2$$

Variables: the curve varies with ϵ_{t-1} while holding $\sigma_{t-1}^2 = \bar{\sigma}^2$.

$$NIC(\epsilon_{t-1}) = \alpha_0 + \beta_1 \frac{\alpha_0}{1 - \alpha_1 - \beta_1} + \alpha_1 \epsilon_{t-1}^2$$

Variables: this substitutes the long-run variance into the NIC.

News impact curve for GJR-GARCH

$$A = \alpha_0 + \beta_1 \frac{\alpha_0}{1 - \alpha_1 - \lambda/2 - \beta_1}$$

Variables: A is the intercept-like NIC component after fixing lagged variance at the GJR long-run variance.

$$\begin{aligned} \epsilon_{t-1} > 0: &\quad \sigma_t^2 = A + \alpha_1 \epsilon_{t-1}^2 \\ \epsilon_{t-1} \le 0: &\quad \sigma_t^2 = A + (\alpha_1 + \lambda) \epsilon_{t-1}^2 \end{aligned}$$

Variables: the two branches show asymmetric curvature for good and bad news.

IGARCH, RiskMetrics, MA, and ARMA

IGARCH persistence

$$\alpha_1 + \beta_1 = 1$$

Variables: IGARCH has unit volatility persistence.

IGARCH(1,1) from GARCH(1,1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_1 = 1 - \beta_1$$

Variables: the second condition enforces $\alpha_1 + \beta_1 = 1$.

IGARCH with zero intercept

$$\alpha_0 = 0$$

Variables: RiskMetrics/EWMA uses zero variance intercept.

$$\sigma_t^2 = (1 - \beta_1) \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: β_1 is the decay weight on old variance.

IGARCH using returns as shocks

$$\sigma_t^2 = (1 - \beta_1) r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Variables: this uses r_{t-1} as the shock when the conditional mean is zero.

IGARCH unconditional variance issue

$$\operatorname{Var}(\epsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \qquad \alpha_1 + \beta_1 = 1$$

Variables: the denominator is zero, so stationary unconditional variance is not finite.

RiskMetrics conditional normal model

$$r_t \mid F_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$$

Variables: RiskMetrics often assumes zero conditional mean.

RiskMetrics variance update

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Variables: λ is the decay factor; $1 - \lambda$ is the latest squared-return weight.

RiskMetrics one-step forecast

$$\sigma_{T+1}^2 = \lambda \sigma_T^2 + (1 - \lambda) r_T^2$$

Variables: r_T is the latest observed return.

Conditional-normal VaR

$$r_{T+1} \mid F_T \sim \mathcal{N}(\mu_{T+1|T}, \sigma_{T+1}^2)$$

Variables: $\mu_{T+1|T}$ is the conditional mean forecast.

$$q_\alpha = \mu_{T+1|T} + \sigma_{T+1} z_\alpha$$

Variables: q_α is the conditional return quantile.

$$\mathrm{VaR}_\alpha = |W_0 q_\alpha|$$

Variables: W_0 is dollar exposure.

$$q_{0.05} = -1.645 \sigma_{T+1}, \qquad \mathrm{VaR}_{0.05} = |W_0(-1.645 \sigma_{T+1})|$$

Variables: this zero-mean form uses $z_{0.05} = -1.645$.

AR-GARCH VaR model

$$\begin{aligned} r_t &= \mu_t + \epsilon_t \\ \mu_t &= c + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p} \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

Variables: μ_t is conditional mean; p is AR order.

AR-GARCH mean forecast

$$\mu_{T+1|T} = c + \phi_1 r_T + \cdots + \phi_p r_{T-p+1}$$

Variables: all returns on the right are known at time T.

AR-GARCH VaR quantile

$$q_{0.05} = \mu_{T+1|T} - 1.645\sigma_{T+1}, \qquad \text{VaR}_{0.05} = |W_0 q_{0.05}|$$

Variables: $\sigma_{T+1} = \sqrt{\sigma^2_{T+1}}$.

Persistence versus volatility level

$$\text{persistence} = \alpha_1 + \beta_1, \qquad \text{long-run variance} = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: persistence controls shock decay; α_0 also affects the volatility level.

MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

Variables: ϕ_0 is mean/intercept; θ_1 is MA coefficient; u_t is shock.

$$u_t \sim \text{iid}(0, \sigma_u^2)$$

Variables: σ_u^2 is innovation variance.

MA(1) mean, variance, and autocorrelation

$$\mathbb{E}(y_t) = \phi_0, \qquad \text{Var}(y_t) = (1 + \theta_1^2)\sigma_u^2$$

Variables: MA(1) variance depends on θ_1^2.

$$\rho_1 = \frac{\theta_1}{1 + \theta_1^2}, \qquad \rho_k = 0 \text{ for } k > 1$$

Variables: MA(1) autocorrelation cuts off after lag 1.

MA(1) forecasts

$$y_{T+1} = \phi_0 + u_{T+1} + \theta_1 u_T$$

Variables: u_T is the latest known/estimated shock.

$$\hat{y}_{T+1} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T, \qquad \hat{y}_{T+h} = \hat{\phi}_0 \text{ for } h \geq 2$$

Variables: future shocks have zero expected value.

ARMA(p,q)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}$$

Variables: p is AR order; q is MA order.

$$\text{lowest AIC} = \text{preferred}$$

Variables: use AIC to choose among candidate ARMA models.

Multi-Step Forecasting and Portfolio VaR

Future squared shocks equal future variance in expectation

$$\mathbb{E}(\epsilon^2_{t+h-1} \mid F_t) = \mathbb{E}(\sigma^2_{t+h-1} \mid F_t)$$

Variables: future squared shocks are unknown, so use their conditional expectation.

$$\epsilon_{t+h-1} = \sigma_{t+h-1} u_{t+h-1}, \qquad \mathbb{E}(u^2_{t+h-1} \mid F_{t+h-2}) = 1$$

Variables: u is standardised to have conditional second moment 1.

Recursive h-step GARCH variance forecast

$$\rho = \alpha_1 + \beta_1$$

Variables: ρ is GARCH persistence.

$$\mathbb{E}(\sigma^2_{t+h} \mid F_t) = \alpha_0 + \rho \mathbb{E}(\sigma^2_{t+h-1} \mid F_t)$$

Variables: the h-step forecast uses the previous horizon's variance forecast.

Closed-form h-step GARCH forecast

$$\begin{aligned}\mathbb{E}(\sigma^2_{t+h} \mid F_t) = \alpha_0 \frac{1 - \rho^{h-1}}{1 - \rho} \\ + \rho^{h-1} \mathbb{E}(\sigma^2_{t+1} \mid F_t)\end{aligned}$$

Variables: h is forecast horizon; ρ^{h-1} controls mean reversion speed.

Closed-form forecast using long-run variance

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Variables: $\bar{\sigma}^2$ is long-run variance.

$$\mathbb{E}(\sigma^2_{t+h} \mid F_t) = \bar{\sigma}^2 + \rho^{h-1} \left[\mathbb{E}(\sigma^2_{t+1} \mid F_t) - \bar{\sigma}^2\right]$$

Variables: forecasts converge to $\bar{\sigma}^2$ when $\rho < 1$.

$$\mathbb{E}(\sigma^2_{t+h} \mid F_t) \to \bar{\sigma}^2$$

Variables: this is the long-horizon mean-reversion result.

Prediction interval for returns

$$r_{T+1} \mid F_T \sim \mathcal{N}(\mu_{T+1|T}, \sigma^2_{T+1})$$

Variables: $\mu_{T+1|T}$ is the one-step mean forecast; σ^2_{T+1} is the one-step variance forecast.

$$\mu_{T+1|T} \pm 1.96\sqrt{\sigma^2_{T+1}}$$

Variables: this is an approximate 95 percent prediction interval.

One-asset VaR with forecast variance

$$q_{0.05} = \mu_{T+1|T} - 1.645\sqrt{\sigma^2_{T+1}}$$

Variables: $q_{0.05}$ is the 5 percent bad-return quantile.

$$\text{VaR}_{0.05} = |W_0 q_{0.05}|$$

Variables: W_0 is position value.

$$\text{VaR}_{0.05} = |W_0(-1.645\sqrt{\sigma^2_{T+1}})|$$

Variables: this is the zero-mean special case.

Asset-specific RiskMetrics forecast

$$\sigma^2_{i,t+1} = (1 - \lambda_i) r^2_{i,t} + \lambda_i \sigma^2_{i,t}$$

Variables: i indexes the asset; λ_i is asset-specific decay factor.

Asset-specific RiskMetrics VaR

$$\text{VaR}_i = \left| W_i(-1.645)\sqrt{\sigma^2_{i,t+1}} \right|$$

Variables: W_i is the dollar exposure to asset i.

Two-asset portfolio return for VaR

$$r_{p,t} = wr_{m,t} + (1-w)r_{c,t}$$

Variables: w is the portfolio weight in asset m; $1-w$ is the weight in asset c.

$$w = \frac{\text{amount in asset } m}{\text{total portfolio value}}, \qquad 1-w = \frac{\text{amount in asset } c}{\text{total portfolio value}}$$

Variables: weights are based on dollar values.

Two-asset portfolio variance for VaR

$$\sigma^2_p = w^2\sigma^2_m + (1-w)^2\sigma^2_c + 2w(1-w)\rho\sigma_m\sigma_c$$

Variables: ρ is correlation between the two asset returns.

Portfolio VaR from portfolio volatility

$$\text{VaR}_p = |\text{total value} \times (-1.645) \times \sigma_p|$$

Variables: $\sigma_p = \sqrt{\sigma^2_p}$ is portfolio standard deviation.

Portfolio VaR from individual VaRs

$$\text{VaR}_p = \sqrt{\text{VaR}^2_m + \text{VaR}^2_c + 2\rho\, \text{VaR}_m\, \text{VaR}_c}$$

Variables: $\text{VaR}_m, \text{VaR}_c$ are stand-alone asset VaRs.

Generic two-asset portfolio variance and VaR notation

$$\sigma^2_p = w^2_A\sigma^2_A + w^2_B\sigma^2_B + 2w_A w_B \rho_{AB}\sigma_A\sigma_B$$

Variables: A, B are generic assets.

$$\text{VaR}_p = W_0(1.645)\sigma_p$$

Variables: this writes VaR as a positive loss number when mean return is zero.

$$\text{VaR}_p = \sqrt{\text{VaR}^2_A + \text{VaR}^2_B + 2\rho_{AB}\, \text{VaR}_A\, \text{VaR}_B}$$

Variables: this is the same normal portfolio VaR logic expressed using individual VaRs.

Essential Exam Answer Templates

ADF template

$$H_0 : \text{unit root / non-stationary}$$
$$H_1 : \text{stationary}$$
$$\text{ADF} = \frac{\text{coefficient}}{\text{standard error}}$$

reject only if ADF is more negative than the critical value

Variables: the coefficient is the ADF coefficient on lagged level.

General t-test template

$$H_0 : \beta = \beta_0$$
$$H_1 : \beta \neq \beta_0$$
$$t = \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})}$$

compare $|t|$ to the critical value

Variables: β_0 is the hypothesised value.

CAPM versus multi-factor template

$$H_0 : \text{all extra factor } \beta\text{s} = 0$$
$$H_1 : \text{at least one extra } \beta \neq 0$$
$$J = \frac{RSS_0 - RSS_1}{RSS_1/(T-K-1)}$$
$$J > \chi^2_{\text{restrictions}} \Rightarrow \text{reject } H_0$$

Variables: RSS_0 is restricted-model RSS; RSS_1 is unrestricted-model RSS.

ARCH test template

estimate mean model

square residuals

regress squared residuals on q lags

$TR^2 \sim \chi^2_q$

reject means ARCH effects / time-varying volatility

Variables: q is number of lagged squared residuals in the auxiliary regression.

One-step GARCH forecast template

$$\text{mean forecast} = \text{fitted mean equation using known latest returns}$$
$$\text{variance forecast} = \alpha_0 + \alpha_1\epsilon^2_T + \beta_1\sigma^2_T$$

Variables: the latest return, shock, and variance are known at time T.

Prediction interval template

$$\text{mean forecast} \pm 1.96\sqrt{\text{variance forecast}}$$

Variables: use 1.96 for an approximate 95 percent interval under normality.

5 percent VaR template

$$q_{0.05} = \text{mean forecast} - 1.645\sqrt{\text{variance forecast}}$$
$$\text{VaR}_{0.05} = |\text{position value} \times q_{0.05}|$$

Variables: 1.645 is the one-sided 5 percent normal cutoff.